Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2010 – Paper 2 – Solutions**

June 2010 – Question 1 – Solution

(a) Find the values of the constant p such that $x - p$ is a factor of

$$f(x) = 4x^3 - (3p + 2)x^2 - (p^2 - 1)x + 3 \qquad [5]$$

$$f(x) = 4x^3 - (3p + 2)x^2 - (p^2 - 1)x + 3$$

Since $(x - p)$ is a factor of $f(x)$, then by the Factor Theorem, $f(p) = 0$.

$$f(p) = 4p^3 - (3p + 2)p^2 - (p^2 - 1)p + 3$$

$$0 = 4p^3 - 3p^3 - 2p^2 - p^3 + p + 3$$

$$0 = -2p^2 + p + 3$$

$$2p^2 - p - 3 = 0$$

$$(2p - 3)(p + 1) = 0$$

Either $\quad 2p - 3 = 0 \quad$ or $\quad p + 1 = 0$

$$2p = 3 \qquad\qquad p = -1$$

$$p = \frac{3}{2}$$

$\therefore$ The values of the constant p are $p = -1$ or $p = \frac{3}{2}$.

(b) Solve, for x and y, the simultaneous equations

$$\log(x-1) + 2\log y = 2\log 3$$

$$\log x + \log y = \log 6$$ [8]

$\log(x-1) + 2\log y = 2\log 3$ → Equation 1

$\log x + \log y = \log 6$ → Equation 2

From Equation 1, we have

$\log(x-1) + 2\log y = 2\log 3$

$\log(x-1) + \log y^2 = \log 3^2$

$\log(x-1)y^2 = \log 9$

Removing logs from both sides gives:

$(x-1)y^2 = 9$

$xy^2 - y^2 = 9$ → Equation 3

From Equation 2, we have

$\log x + \log y = \log 6$

$\log xy = \log 6$

Removing logs from both sides gives:

$xy = 6$

$x = \frac{6}{y}$ → Equation 4

Substituting Equation 4 into Equation 3 gives:

$$\left(\frac{6}{y}\right)y^2 - y^2 = 9$$

$$6y - y^2 = 9$$

$$y^2 - 6y + 9 = 0$$

$$(y-3)^2 = 0$$

$$y - 3 = 0$$

$$y = 3$$

Substituting $y = 3$ into Equation 4 gives:

$$x = \frac{6}{3}$$

$$= 2$$

$\therefore x = 2$ and $y = 3$

(c) Solve, for $x \in \mathbb{R}$, the inequality

$$\frac{2x-3}{x+1} - 5 > 0 \qquad [5]$$

$$\frac{2x-3}{x+1} - 5 > 0$$

Multiplying throughout by $(x+1)^2$ gives:

$$(2x-3)(x+1) - 5(x+1)^2 > 0$$

$$2x^2 - 3x + 2x - 3 - 5(x^2 + 2x + 1) > 0$$

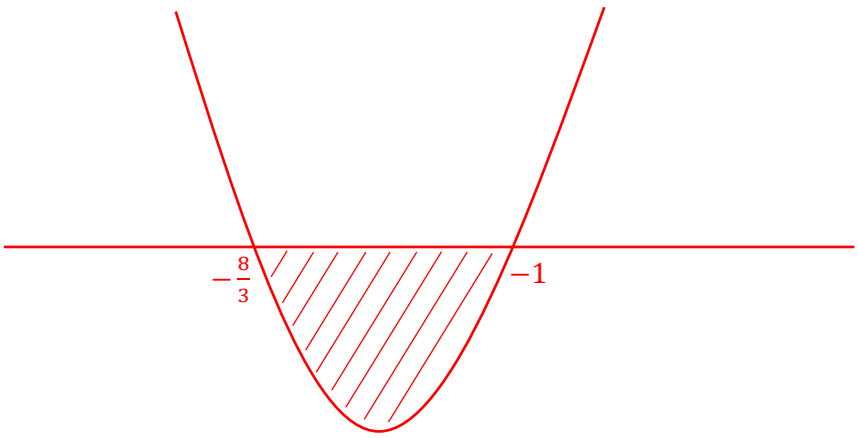
$$2x^2 - x - 3 - 5x^2 - 10x - 5 > 0$$

$$-3x^2 - 11x - 8 > 0$$

$$3x^2 + 11x + 8 < 0$$

$$(3x+8)(x+1) < 0$$

The quadratic curve cuts the x-axis at $x = -\frac{8}{3}$ and $x = -1$.



We are looking at the area where $(3x + 8)(x + 1) < 0$.

$\therefore$ The solution set is $\left\{x: -\frac{8}{3} < x < -1\right\}$.

(d) By using $y = 2^x$, or otherwise, solve

$$4^x - 3(2^{x+1}) + 8 = 0$$ [7]

$$4^x - 3(2^{x+1}) + 8 = 0$$

$$(2^2)^x - 3(2^x)(2) + 8 = 0$$

$$(2^x)^2 - 6(2^x) + 8 = 0$$

Let $y = 2^x$.

Then, we have,

$$y^2 - 6y + 8 = 0$$

$$(y - 4)(y - 2) = 0$$

Either	$y - 4 = 0$	or	$y - 2 = 0$
	$y = 4$		$y = 2$
	$2^x = 4$		$2^x = 2$
	$2^x = 2^2$		$2^x = 2^1$
	$x = 2$		$x = 1$

$\therefore x = 1$ or $x = 2$

Total: 25 marks

June 2010 – Question 2 – Solution

(a) (i) Use the fact that $S_n = \sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$ to express

$S_{2n} = \sum_{r=1}^{2n} r$ in terms of n. [2]

$S_n = \sum_{r=1}^{n} r$ and $\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$

Now,

$S_{2n} = \sum_{r=1}^{2n} r$

$= \frac{1}{2}(2n)(2n+1)$

$= n(2n+1)$

$\therefore S_{2n} = n(2n+1)$

(ii) Find constants p and q such that

$S_{2n} - S_n = pn^2 + qn$ [5]

$S_{2n} - S_n = n(2n+1) + \frac{1}{2}n(n+1)$

$= 2n^2 + n - \frac{1}{2}n^2 - \frac{1}{2}n$

$= \frac{3}{2}n^2 + \frac{1}{2}n$

We are given that $S_{2n} - S_n = pn^2 + qn$.

Comparing and equating coefficients gives:

$p = \frac{3}{2}$ and $q = \frac{1}{2}$

(iii) Hence, or otherwise, find n such that

$$S_{2n} - S_n = 260 \qquad [5]$$

$S_{2n} - S_n = 260$

$S_{2n} - S_n = \frac{3}{2}n^2 + \frac{1}{2}n$

Equating both equations gives:

$$\frac{3}{2}n^2 + \frac{1}{2}n = 260$$

$$3n^2 + n = 520$$

$$3n^2 + n - 520 = 0$$

$$(3n + 40)(n - 13) = 0$$

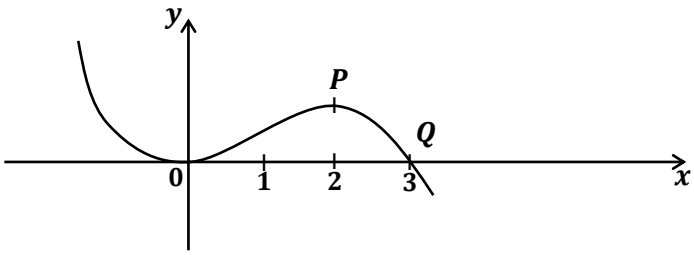
Either $3n + 40 = 0$ or $n - 13 = 0$

$3n = -40$ $\qquad$ $n = 13$

$n = -\frac{40}{3}$

Since n cannot be negative, then $n = 13$.

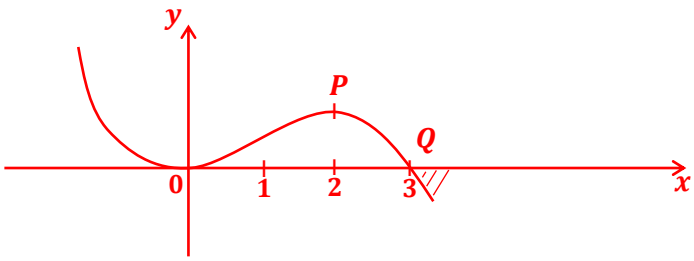
(b) The diagram below (**not drawn to scale**) shows the graph of $y = x^2(3 - x)$. The coordinates of points P and Q are (2,4) and (3,0) respectively.



(i) Write down the solution set of the inequality $x^2(3 - x) \leq 0$. [4]

$x^2(3 - x) \leq 0$

The graph cuts the x-axis at $x = 0$ and $x = 3$.



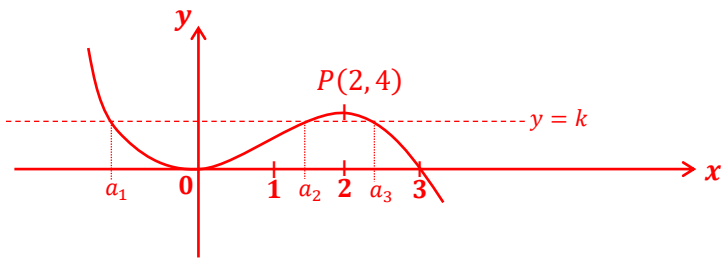
From the diagram, the graph is 0 at the origin and at $x = 3$ and negative for values of x greater than 3.

$\therefore$ The solution set is $\{x: x = 0 \cup x \geq 3\}$.

(ii) Given that the equation $x^2(3 - x) = k$ has three real solutions for x, write down the set of possible values for k. [3]

$x^2(3 - x) = k$

Consider the diagram below:



There are three real solutions, a_1, a_2 and a_3, for $y = k$.

For three real solutions, $k < 4$ and $k > 0$.

$\therefore$ The possible set of values for k are $\{k: 0 < k < 4\}$.

(iii) The functions f and g are defined as follows:

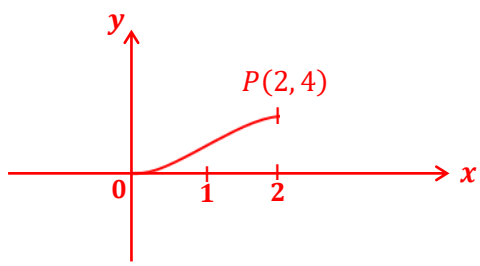
$f: x \to x^2(3 - x), \qquad 0 < x < 2$

$g: x \to x^2(3 - x), \qquad 0 < x < 3$

By using (b)(ii) above, or otherwise, show that

(a) f has an inverse

$f(x) = x^2(3 - x), \qquad 0 < x < 2$



Any line drawn parallel to the x-axis will cut the graph of $f(x)$ at most once. So, $f(x)$ is injective.

Any line drawn parallel to the x-axis will cut the graph of $f(x)$ at least once. So, $f(x)$ is surjective.

Since $f(x)$ is both injective and surjective, then $f(x)$ is bijective.

The inverse of a function exists if and only if the function is bijective.

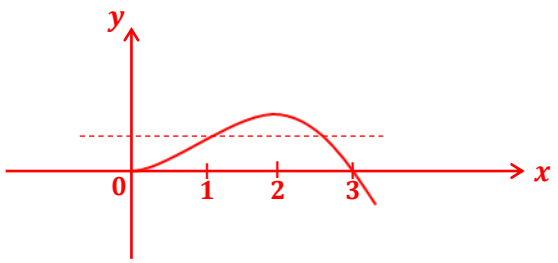
$\therefore$ $f(x)$ is bijective and the function f has an inverse.

Q.E.D.

(b) g does NOT have an inverse [6]

$g(x) = x^2(3 - x), \quad 0 < x < 3$

Consider the diagram below:



Any line drawn parallel to the x-axis will cut the graph more than once.

Hence, $g(x)$ is not injective. This implies that $g(x)$ is not bijective.

$\therefore$ g does not have an inverse.

Q.E.D.

Total: 25 marks

June 2010 – Question 3 – Solution

(a) The vectors $\boldsymbol{p}$ and $\boldsymbol{q}$ are given by

$$\boldsymbol{p} = 6\hat{\boldsymbol{\imath}} + 4\hat{\boldsymbol{\jmath}}$$

$$\boldsymbol{q} = -8\hat{\boldsymbol{\imath}} - 9\hat{\boldsymbol{\jmath}}$$

(i) Calculate, in degrees, the angle between $\boldsymbol{p}$ and $\boldsymbol{q}$. [5]

$$\boldsymbol{p} \cdot \boldsymbol{q} = \begin{pmatrix} 6 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} -8 \\ -9 \end{pmatrix}$$

$$= (6)(-8) + (4)(-9)$$

$$= -48 - 36$$

$$= -84$$

$$|\boldsymbol{p}| = \sqrt{(6)^2 + (4)^2}$$

$$= \sqrt{36 + 16}$$

$$= \sqrt{52}$$

$$|\boldsymbol{q}| = \sqrt{(-8)^2 + (-9)^2}$$

$$= \sqrt{64 + 81}$$

$$= \sqrt{145}$$

Now,

$$\cos\theta = \frac{\boldsymbol{p} \cdot \boldsymbol{q}}{|\boldsymbol{p}||\boldsymbol{q}|}$$

$$\cos\theta = \frac{-84}{\sqrt{52}\sqrt{145}}$$

$$\theta = \cos^{-1}\left(\frac{-84}{\sqrt{52}\sqrt{145}}\right)$$

$$\theta = 165.3° \quad \text{(to 1 decimal place)}$$

(ii) (a) Find a non-zero vector $\boldsymbol{v}$ such that $\boldsymbol{p} \cdot \boldsymbol{v} = 0$.

Let $\boldsymbol{v} = a\hat{\boldsymbol{\imath}} + b\hat{\boldsymbol{\jmath}}$, where $a, b \in \mathbb{Z}$.

We have $\boldsymbol{p} = 6\hat{\boldsymbol{\imath}} + 4\hat{\boldsymbol{\jmath}}$.

$$\boldsymbol{p} \cdot \boldsymbol{v} = 0$$

$$(6\hat{\boldsymbol{\imath}} + 4\hat{\boldsymbol{\jmath}})(a\hat{\boldsymbol{\imath}} + b\hat{\boldsymbol{\jmath}}) = 0$$

$$\begin{pmatrix} 6 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \end{pmatrix} = 0$$

$$6a + 4b = 0$$

$$3a + 2b = 0$$

There exists any number of possible values of a and b where $3a + 2b = 0$.

Suppose $a = -2$ and $b = 3$.

Then,

$$3(-2) + 2(3) = -6 + 6$$

$$= 0$$

$\therefore$ A non-zero vector $\boldsymbol{v}$ such that $\boldsymbol{p} \cdot \boldsymbol{v} = 0$ is $\boldsymbol{v} = -2\hat{\boldsymbol{\imath}} + 3\hat{\boldsymbol{\jmath}}$.

(b) State the relationship between $\boldsymbol{p}$ and $\boldsymbol{v}$. [5]

Since $\boldsymbol{p} \cdot \boldsymbol{v} = 0$, then $\boldsymbol{p}$ and $\boldsymbol{v}$ are perpendicular to each other.

(b) The circle C_1 has $(-3, 4)$ and $(1, 2)$ as endpoints of a diameter.

(i) Show that the equation of C_1 is $x^2 + y^2 + 2x - 6y + 5 = 0$. [6]

The endpoints of a diameter are $(-3, 4)$ and $(1, 2)$.

The centre of the diameter is the midpoint of the endpoints.

$$\text{Midpoint} = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$$

$$= \left(\frac{-3+1}{2}, \frac{4+2}{2}\right)$$

$$= \left(\frac{-2}{2}, \frac{6}{2}\right)$$

$$= (-1, 3)$$

So, the centre of the circle is $(-1, 3)$.

The radius of the circle will be equal to the length between $(-1, 3)$ and $(1, 2)$.

$$\text{Length} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(-1-1)^2 + (3-2)^2}$$

$$= \sqrt{(-2)^2 + (1)^2}$$

$$= \sqrt{4+1}$$

$$= \sqrt{5} \text{ units}$$

So, the radius of the circle is $\sqrt{5}$ units.

The standard form for the equation of a circle is:

$$(x-a)^2 + (y-b)^2 = r^2$$

Substituting centre $(-1, 3)$ and radius, $r = \sqrt{5}$ units gives:

$$\left(x - (-1)\right)^2 + (y - 3)^2 = \left(\sqrt{5}\right)^2$$

$$(x + 1)^2 + (y - 3)^2 = 5$$

$$x^2 + 2x + 1 + y^2 - 6y + 9 = 5$$

$$x^2 + 2x + y^2 - 6y + 10 = 5$$

$$x^2 + y^2 + 2x - 6y + 10 - 5 = 0$$

$$x^2 + y^2 + 2x - 6y + 5 = 0$$

$\therefore$ The equation of C_1 is $x^2 + y^2 + 2x - 6y + 5 = 0$.

Q.E.D.

(ii) The circle C_2 has equation $x^2 + y^2 + x - 5y = 0$. Calculate the coordinates of the points of intersection of C_1 and C_2. [9]

C_1: $x^2 + y^2 + 2x - 6y + 5 = 0$ $\rightarrow$ Equation 1

C_2: $x^2 + y^2 + x - 5y = 0$ $\rightarrow$ Equation 2

Equating both equations gives:

$$x^2 + y^2 + 2x - 6y + 5 = x^2 + y^2 + x - 5y$$

$$2x - 6y + 5 = x - 5y$$

$$2x - x + 5 = 6y - 5y$$

$$x + 5 = y$$

$$y = x + 5$$

Substituting $y = x + 5$ into Equation 2 gives:

$$x^2 + (x+5)^2 + x - 5(x+5) = 0$$

$$x^2 + x^2 + 10x + 25 + x - 5x - 25 = 0$$

$$2x^2 + 6x = 0$$

$$x^2 + 3x = 0$$

$$x(x+3) = 0$$

Either $x = 0$ or $x + 3 = 0$

$x = -3$

When $x = 0$,

$y = (0) + 5$

$= 5$

When $x = -3$,

$y = (-3) + 5$

$= 2$

$\therefore$ C_1 and C_2 intersect at the points $(0, 5)$ and $(-3, 2)$.

Total: 25 marks

June 2010 – Question 4 – Solution

(a) (i) Solve the equation $\cos 3A = 0.5$ for $0 \leq A \leq \pi$. [4]

Required to solve the equation $\cos 3A = 0.5$ for $0 \leq A \leq \pi$.

Well, $\cos 3A = \frac{1}{2}$

$PV = \frac{\pi}{3}$

G.S. is $3A = 2n\pi \pm PV$

$3A = 2n\pi \pm \frac{\pi}{3}$

$3A = \frac{\pi}{3}, 2\pi - \frac{\pi}{3}, 2\pi + \frac{\pi}{3}, \ldots$

$3A = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \ldots$

$A = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$ for $0 \leq A \leq \pi$.

The other values that satisfy the equation are not within the given range.

$\therefore A = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$ for $0 \leq A \leq \pi$.

(ii) Show that $\cos 3A = 4\cos^3 A - 3\cos A$. [6]

Taking L.H.S:

$\cos 3A = \cos(2A + A)$

$= \cos 2A \cos A - \sin 2A \sin A$

$= (2\cos^2 A - 1)(\cos A) - (2\sin A \cos A)\sin A$

$= 2\cos^3 A - \cos A - 2\sin^2 A \cos A$

$= 2\cos^3 A - \cos A - 2(1 - \cos^2 A)\cos A$

$= 2\cos^3 A - \cos A - 2\cos A + 2\cos^3 A$

$= 4\cos^3 A - 3\cos A$

$=$ R.H.S.

$\therefore \cos 3A = 4\cos^3 A - 3\cos A.$

Q.E.D.

(iii) The THREE roots of the equation $4p^3 - 3p - 0.5 = 0$ all lie between -1 and 1.

Use the results in (a)(i) and (ii) to find these roots. [4]

From (a)(i), we have $\cos 3A = 0.5$.

From (a)(ii), we have $\cos 3A = 4\cos^3 A - 3\cos A$.

Equating gives,

$4\cos^3 A - 3\cos A = 0.5$

$4\cos^3 A - 3\cos A - 0.5 = 0$

Let $p = \cos A$.

Then,

$4p^3 - 3p - 0.5 = 0$

From (a)(i), the solutions of A were

$A = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$

When $A = \frac{\pi}{9}$,

$p = \cos\left(\frac{\pi}{9}\right)$

$= 0.940$

When $A = \frac{5\pi}{9}$,

$p = \cos\left(\frac{5\pi}{9}\right)$

$= -0.174$

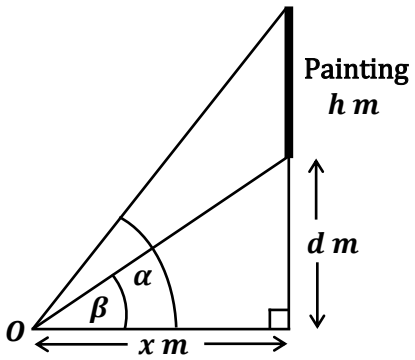
When $A = \frac{7\pi}{9}$,

$p = \cos\left(\frac{7\pi}{9}\right)$

$= -0.766$

$\therefore$ The roots of the equation are $p = -0.766, -0.174, 0.940$ for $-1 \leq p \leq 1$.

(b) The following diagram, not drawn to scale, represents a painting of height, h metres, that is fastened to a vertical wall at a height of d metres above, and x metres away from, the level of an observer, O.



The viewing angle of the painting is $(\alpha - \beta)$, where α and β are respectively the angles of inclination, in radians, from the level of the observer to the top and base of the painting.

(i) Show that $\tan(\alpha - \beta) = \frac{hx}{x^2+d(d+h)}$. [6]

$\tan\alpha = \frac{opp}{adj}$

$= \frac{h+d}{x}$

$\tan\beta = \frac{opp}{adj}$

$= \frac{d}{x}$

Now,

$\tan(\alpha - \beta) = \frac{\tan\alpha - \tan\beta}{1 - \tan\alpha\tan\beta}$

Substituting $\tan\alpha = \frac{h+d}{x}$ and $\tan\beta = \frac{d}{x}$ into $\tan(\alpha-\beta) = \frac{\tan\alpha - \tan\beta}{1-\tan\alpha\tan\beta}$ gives:

$$\tan(\alpha-\beta) = \frac{\left(\frac{h+d}{x}\right)-\left(\frac{d}{x}\right)}{1+\left(\frac{h+d}{x}\right)\left(\frac{d}{x}\right)}$$

$$= \frac{\frac{h+d-d}{x}}{1+\frac{d(d+h)}{x^2}}$$

$$= \frac{\frac{h}{x}}{1+\frac{d(d+h)}{x^2}}$$

$$= \frac{h}{x} \div \left(1+\frac{d(d+h)}{x^2}\right)$$

$$= \frac{h}{x} \div \left(\frac{x^2}{x^2}+\frac{d(d+h)}{x^2}\right)$$

$$= \frac{h}{x} \div \frac{x^2+d(d+h)}{x^2}$$

$$= \frac{h}{x} \times \frac{x^2}{x^2+d(d+h)}$$

$$= \frac{hx}{x^2+d(d+h)}$$

$$= \text{R.H.S.}$$

$$\therefore \tan(\alpha-\beta) = \frac{hx}{x^2+d(d+h)}$$

Q.E.D.

(ii) The viewing angle of the painting, $(\alpha-\beta)$, is at a maximum where $x = \sqrt{h(d+h)}$. Calculate the maximum viewing angle, in radians, when $d = 3h$. [5]

$$\tan(\alpha-\beta) = \frac{hx}{x^2+d(d+h)}$$

Substituting $x = \sqrt{h(d+h)}$ and $d = 3h$ gives:

$$\tan(\alpha-\beta)=\frac{h\left(\sqrt{h(3h+h)}\right)}{\left(\sqrt{h(3h+h)}\right)^2+3h(3h+h)}$$

$$\tan(\alpha-\beta)=\frac{h\left(\sqrt{3h^2+h^2}\right)}{\left(\sqrt{3h^2+h^2}\right)^2+(9h^2+3h^2)}$$

$$\tan(\alpha-\beta)=\frac{h\left(\sqrt{4h^2}\right)}{\left(\sqrt{4h^2}\right)^2+12h^2}$$

$$\tan(\alpha-\beta)=\frac{h(2h)}{4h^2+12h^2}$$

$$\tan(\alpha-\beta)=\frac{2h^2}{16h^2}$$

$$\tan(\alpha-\beta)=\frac{1}{8}$$

$$\alpha-\beta=\tan^{-1}\left(\frac{1}{8}\right)$$

$$\alpha-\beta=0.124^c$$

$\therefore$ The maximum viewing angle is 0.124^c.

Total: 25 marks

June 2010 – Question 5 – Solution

(a) Find

(i) $\lim_{x\to 3} \frac{x^2-9}{x^3-27}$ [4]

$$\lim_{x\to 3} \frac{x^2-9}{x^3-27} = \lim_{x\to 3} \frac{(x-3)(x+3)}{(x-3)(x^2+3x+9)}$$

$$= \lim_{x\to 3} \frac{x+3}{x^2+3x+9}$$

$$= \frac{(3)+3}{(3)^2+3(3)+9}$$

$$= \frac{6}{9+9+9}$$

$$= \frac{6}{27}$$

$$= \frac{2}{9}$$

(ii) $\lim_{x\to 0} \frac{\tan x-5x}{\sin 2x-4x}$ [5]

$$\lim_{x\to 0} \frac{\tan x-5x}{\sin 2x-4x} = \lim_{x\to 0} \left(\frac{\tan x-5x}{x} \cdot \frac{x}{\sin 2x-4x}\right)$$

$$= \lim_{x\to 0} \left(\frac{\tan x-5x}{x} \div \frac{\sin 2x-4x}{x}\right)$$

$$= \lim_{x\to 0} \frac{\tan x-5x}{x} \div \lim_{x\to 0} \frac{\sin 2x-4x}{x}$$

Consider $\lim_{x\to 0} \frac{\tan x-5x}{x}$.

$$\lim_{x\to 0} \frac{\tan x-5x}{x} = \lim_{x\to 0} \left(\frac{\tan x}{x} - 5\right)$$

$$= \lim_{x\to 0} \frac{\tan x}{x} - \lim_{x\to 0} 5$$

$$= 1 - 5$$

$$= -4$$

Consider $\lim_{x\to 0} \frac{x}{\sin 2x - 4x}$.

$$\lim_{x\to 0} \frac{\sin 2x - 4x}{x} = \lim_{x\to 0} \left(\frac{\sin 2x}{x} - 4\right)$$

$$= \lim_{x\to 0} \frac{\sin 2x}{x} - \lim_{x\to 0} 4$$

$$= \lim_{x\to 0} \frac{\sin 2x}{x} \cdot \frac{2}{2} - \lim_{x\to 0} 4$$

$$= 2 \lim_{x\to 0} \frac{\sin 2x}{2x} - \lim_{x\to 0} 4$$

$$= 2(1) - 4$$

$$= 2 - 4$$

$$= -2$$

Now,

$$\lim_{x\to 0} \frac{\tan x - 5x}{\sin 2x - 4x} = \lim_{x\to 0} \frac{\tan x - 5x}{x} \div \lim_{x\to 0} \frac{\sin 2x - 4x}{x}$$

$$= (-4) \div (-2)$$

$$= 2$$

(b) The function f on $\mathbb{R}$ is defined by

$$f(x) = \begin{cases} 3x - 7, & if\ x > 4 \\ 1 + 2x, & if\ x \leq 4 \end{cases}$$

(i) Find

(a) $\lim_{x \to 4^+} f(x)$ [2]

$$\lim_{x \to 4^+} f(x) = \lim_{x \to 4^+} (3x - 7)$$

$$= 3(4) - 7$$

$$= 12 - 7$$

$$= 5$$

(b) $\lim_{x \to 4^-} f(x)$ [2]

$$\lim_{x \to 4^-} f(x) = \lim_{x \to 4^-} (1 + 2x)$$

$$= 1 + 2(4)$$

$$= 1 + 8$$

$$= 9$$

(ii) Deduce that $f(x)$ is discontinuous at $x = 4$. [2]

Since $\lim_{x \to 4^+} f(x) = 5$ and $\lim_{x \to 4^-} f(x) = 9$, then $\lim_{x \to 4^+} f(x) \neq \lim_{x \to 4^-} f(x)$.

Since $\lim_{x \to 4^+} f(x) \neq \lim_{x \to 4^-} f(x)$, then $\lim_{x \to 4} f(x)$ does not exist.

Hence, $f(x)$ is discontinuous at $x = 4$.

(c) (i) Evaluate $\int_{-1}^{1}\left(x-\frac{1}{x}\right)^2 dx$. [6]

$$\int_{-1}^{1}\left(x-\frac{1}{x}\right)^2 dx = \int_{-1}^{1}\left(x^2-2+\frac{1}{x^2}\right)dx$$

$$= \int_{-1}^{1}(x^2-2+x^{-2})\,dx$$

$$= \left[\frac{x^3}{3}-2x-x^{-1}\right]_{-1}^{1}$$

$$= \left[\frac{(1)^3}{3}-2(1)-(1)^{-1}\right]-\left[\frac{(-1)^3}{3}-2(-1)-(-1)^{-1}\right]$$

$$= \left(\frac{1}{3}-2-1\right)-\left(-\frac{1}{3}+2-(-1)\right)$$

$$= -\frac{8}{3}-\left(\frac{8}{3}\right)$$

$$= -\frac{16}{3}$$

$$\therefore \int_{-1}^{1}\left(x-\frac{1}{x}\right)^2 dx = -\frac{16}{3}$$

(ii) Using the substitution $u = x^2+4$, or otherwise, find

$$\int x\sqrt{x^2+4}\;dx$$ [4]

Let $u = x^2+4$

$$\frac{du}{dx} = 2x$$

$$dx = \frac{du}{2x}$$

Now,

$$\int x\sqrt{x^2+4}\ dx = \int x\sqrt{u}\ \frac{du}{2x}$$

$$= \frac{1}{2}\int u^{\frac{1}{2}}\ du$$

$$= \frac{1}{2}\left[\frac{u^{\frac{3}{2}}}{\frac{3}{2}}\right] + c$$

$$= \left(\frac{1}{2}\right)\left(\frac{2}{3}\right)\left[u^{\frac{3}{2}}\right] + c$$

$$= \frac{1}{3}u^{\frac{3}{2}} + c$$

$$= \frac{1}{3}(x^2+4)^{\frac{3}{2}} + c$$

$$\therefore \int x\sqrt{x^2+4}\ dx = \frac{1}{3}(x^2+4)^{\frac{3}{2}} + c$$

Total: 25 marks

June 2010 – Question 6 – Solution

(a) Differentiate with respect to x

(i) $y = \sin(3x + 2) + \tan 5x$ [3]

Let $a = \sin(3x + 2)$

Let $t = 3x + 2$, $a = \sin t$

$\frac{dt}{dx} = 3$ $\frac{da}{dt} = \cos t$

So,

$\frac{da}{dx} = \frac{da}{dt} \times \frac{dt}{dx}$

$= \cos t \times 3$

$= 3\cos(3x + 2)$

Now, let $b = \tan 5x$

Let $t = 5x$, $b = \tan t$

$\frac{dt}{dx} = 5$ $\frac{db}{dt} = \sec^2 t$

So,

$\frac{db}{dx} = \frac{db}{dt} \times \frac{dt}{dx}$

$= \sec^2 t \times 5$

$= 5\sec^2 5x$

$\therefore \frac{dy}{dx} = 3\cos(3x + 2) + 5\sec^2 5x$

(ii) $y = \frac{x^2+1}{x^3-1}$ [4]

$y = \frac{x^2+1}{x^3-1}$ which is of the form $y = \frac{u}{v}$

Let $u = x^2 + 1$, $v = x^3 - 1$

$\frac{du}{dx} = 2x$ $\frac{dv}{dx} = 3x^2$

Using quotient rule,

$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

$$= \frac{(x^3-1)(2x)-(x^2+1)(3x^2)}{(x^3-1)^2}$$

$$= \frac{2x^4-2x-3x^4-3x^2}{(x^3-1)^2}$$

$$= \frac{-x^4-3x^2-2x}{(x^3-1)^2}$$

(b) The function $f(x)$ satisfies $\int_1^4 f(x)\,dx = 7$.

(i) Find $\int_1^4 [3f(x) + 4]\,dx$. [4]

$$\int_1^4 [3f(x) + 4]\,dx = 3\int_1^4 f(x)\,dx + \int_1^4 4\,dx$$

$$= 3(7) + [4x]_1^4$$

$$= 21 + 4(4 - 1)$$

$$= 21 + 4(3)$$

$$= 21 + 12$$

$$= 33$$

(ii) Using the substitution $u = x + 1$, evaluate $\int_0^3 2f(x+1)\,dx$. [4]

Let $u = x + 1$

$\frac{du}{dx} = 1$

$dx = du$

When $x = 0$,

$u = (0) + 1$

$= 1$

When $x = 3$

$u = (3) + 1$

$= 4$

$\int_0^3 2f(x+1)\,dx = \int_1^4 2f(u)\,du$

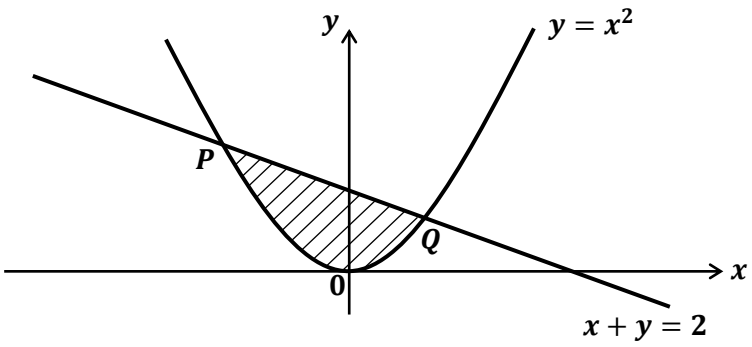
$= 2\int_1^4 f(u)\,du$

$= 2(7)$

$= 14$

$\therefore \int_0^3 2f(x+1)\,dx = 14$

(c) In the diagram below (**not drawn to scale**), the line $x + y = 2$ intersects the curve $y = x^2$ at the points P and Q.



(i) Find the coordinates of the points P and Q. [5]

$y = x^2$ → Equation 1

$x + y = 2$ → Equation 2

Substituting Equation 1 into Equation 2 gives:

$$x + x^2 = 2$$

$$x^2 + x - 2 = 0$$

$$(x - 1)(x + 2) = 0$$

Either $x - 1 = 0$ or $x + 2 = 0$

$x = 1$ $x = -2$

When $x = 1$,

$$y = (1)^2$$

$$= 1$$

When $x = -2$,

$$y = (-2)^2$$

$$= 4$$

$\therefore$ The coordinates of the points are $P(-2, 4)$ and $Q(1, 1)$.

(ii) Calculate the area of the shaded portion of the diagram bounded by the curve and the straight line. [5]

The equation of the line is:

$x + y = 2$

$$y = 2 - x$$

Area under the line $= \int_a^b y \ dx$

$$= \int_{-2}^{1} (2 - x)\, dx$$

$$= \left[2x - \frac{x^2}{2}\right]_{-2}^{1}$$

$$= \left[2(1) - \frac{(1)^2}{2}\right] - \left[2(-2) - \frac{(-2)^2}{2}\right]$$

$$= \left(2 - \frac{1}{2}\right) - (-4 - 2)$$

$$= \frac{3}{2} - (-6)$$

$$= \frac{15}{2}\ units^2$$

The equation of the curve is:

$y = x^2$

Area under curve $= \int_a^b y \ dx$

$= \int_{-2}^{1} x^2 \ dx$

$= \left[\frac{x^3}{3}\right]_{-2}^{1}$

$= \left[\frac{(1)^3}{3}\right] - \left[\frac{(-2)^3}{3}\right]$

$= \left(\frac{1}{3}\right) - \left(-\frac{8}{3}\right)$

$= \frac{1}{3} + \frac{8}{3}$

$= \frac{9}{3}$

$= 3 \ units^2$

Hence,

Area of shaded region = Area under the line − Area under curve

$= \frac{15}{2} - 3$

$= \frac{9}{2} \ units^2$

∴ The area of the shaded portion of the diagram bounded by the curve and the straight line is $\frac{9}{2} \ units^2$.

Total: 25 marks